

gsDesignNB: Group Sequential Design and Sample Size Re-estimation for Negative Binomial Recurrent Event Trials

Abstract

We present the R package **gsDesignNB** for planning, monitoring, and adapting clinical trials with recurrent event endpoints analyzed using negative binomial models. In the working model, conditional on treatment arm and a subject-level gamma frailty, recurrent events follow a Poisson process with constant within-subject rate, yielding marginal negative binomial counts. The package implements sample size calculations that extend the methods of Zhu and Lakkis (2014) and Friede and Schmidli (2010) to handle piecewise accrual, exponential dropout, maximum follow-up truncation, and event gaps. Because statistical information for the log rate ratio depends jointly on exposure, event rates, and dispersion, it is not directly proportional to raw event counts. We derive and implement a second-order Taylor correction for Jensen’s inequality bias in the event gap effective rate; this correction was needed for simulation-based power to agree with the planning formula when event gaps were non-negligible, reducing representative approximation error from 8% to 1%. A 64-scenario simulation sweep across clinically relevant event rates, dispersion values, and gap durations confirms that the correction prevents 1–2.5 percentage points of underpowering when dispersion is moderate-to-high ($k \geq 0.5$) and gaps exceed 20 days, while adding negligibly to sample size in low-dispersion settings. The package supports group sequential designs via integration with the **gsDesign** package, and provides both blinded and unblinded sample size re-estimation at interim analyses. A simulation study across 13 scenarios demonstrates that sample size re-estimation effectively recovers power when nuisance parameters are misspecified, provided the treatment effect is as planned. In the vignette design used here, null simulations suggested slight Type I inflation at nominal one-sided ($= 0.025$); rebuilding at a smaller nominal alpha improved calibration, and blinded updates generally requested larger adapted sample sizes than unblinded updates.

1 Introduction

Recurrent event endpoints are common in clinical trials for conditions such as multiple sclerosis relapses, COPD exacerbations, and heart failure hospitalizations. The negative binomial (NB) model provides a natural framework for analyzing such data, accounting for both within-subject correlation (overdispersion) and variable follow-up through an exposure offset (Friede and Schmidli 2010; Zhu and Lakkis 2014).

Sample size calculation for NB outcomes requires specifying not only the event rates and treatment effect but also the dispersion parameter, average exposure time, and potentially event gap durations. In practice, several of these parameters may be uncertain at the design stage. Group sequential designs allow for interim monitoring, and sample size re-estimation (SSR) can adapt the trial size when nuisance parameters deviate from assumptions. For NB recurrent-event designs, the design target is the statistical information for the log rate ratio, which depends jointly on exposure, event rates, and dispersion; raw event counts are only an incomplete proxy.

The **gsDesignNB** package for R provides an integrated toolkit for these tasks:

1. Sample size and power for fixed designs with flexible enrollment, dropout, follow-up truncation, and event gaps.
2. Group sequential designs via calendar-time analysis schedules compatible with the **gsDesign** package (Jennison and Turnbull 1999).
3. Blinded and unblinded SSR at interim analyses.
4. Simulation utilities for validating designs and evaluating operating characteristics.

This paper describes the methodology, highlights a novel correction for Jensen’s inequality bias in the event gap formula, and presents a simulation study of SSR across multiple scenarios. A brief final section discusses AI-assisted development as implementation support rather than as the primary methodological contribution.

2 Sample size methodology

2.1 Negative binomial model

For subject i in treatment group g with exposure time t_i , the event count Y_i follows a negative binomial distribution with mean $\mu_i = \lambda_g t_i$ and variance $\text{Var}(Y_i) = \mu_i + k\mu_i^2$, where λ_g is the group-specific event rate and k is the dispersion parameter. This arises from a Gamma–Poisson mixture: each subject has a latent rate $\Lambda_i \sim \text{Gamma}(1/k, k\lambda_g)$ and generates events from $\text{Poisson}(\Lambda_i t_i)$.

2.2 Wald test and sample size formula

The treatment effect is the log rate ratio $\theta = \log(\lambda_2/\lambda_1)$, estimated via maximum likelihood under a negative binomial GLM with log link and exposure offset. The design target is statistical information, $\mathcal{J} = 1/\text{Var}(\hat{\theta})$, not the raw event count alone. The Fisher information per subject is $\mathcal{J}_i = \mu_i/(1 + k\mu_i)$. If $W_g = \sum_{i \in g} \mu_i/(1 + k\mu_i)$, then

$$\mathcal{J} = \frac{1}{1/W_1 + 1/W_2}.$$

When exposures are summarized by arm-specific averages and the randomization ratio is $r = n_2/n_1$, this leads to the fixed-design sample size formula

$$n_1 = \frac{(z_{\alpha/s} + z_{\beta})^2}{(\theta - \theta_0)^2} \left[\left(\frac{1}{\mu_1} + k_1 \right) + \frac{1}{r} \left(\frac{1}{\mu_2} + k_2 \right) \right]$$

where $\mu_g = \lambda_g \bar{t}_g$, $\bar{t}_g$ is the average exposure, and k_g may be inflated by a variance correction factor $Q_g = E[t_g^2]/(E[t_g])^2$ to account for variable follow-up (Zhu and Lakkis 2014). Accordingly, two analyses with similar event totals can carry different information if follow-up or dispersion differs.

This formula is equivalent across the three primary references: Method 3 of Zhu and Lakkis (2014) (fixed sample size), Friede and Schmidli (2010) (blinded SSR), and Mütze et al. (2019) (group sequential designs). The practical difference from more familiar first-event formulas is that when the protocol imposes an event gap, the expected count μ_g must reflect the gap-corrected effective rate rather than a naive event-count proxy.

2.3 Jensen’s inequality correction for event gaps

In trials with a mandatory gap of duration γ after each event (e.g., a recovery period), the effective rate is commonly approximated as $\lambda_{\text{eff}} = \lambda/(1 + \lambda\gamma)$. However, this is a population-level average that ignores subject-level heterogeneity. Since $f(x) = x/(1 + x\gamma)$ is concave and subject rates Λ_i are random (Gamma-distributed), Jensen’s inequality gives $E[f(\Lambda)] < f(E[\Lambda])$.

We derive a second-order Taylor correction:

$$\lambda_{\text{eff}} \approx \frac{\lambda}{1 + \lambda\gamma} \left(1 - \frac{k\lambda\gamma}{(1 + \lambda\gamma)^2} \right)$$

using $f''(\lambda) = -2\gamma/(1 + \lambda\gamma)^3$ and $\text{Var}(\Lambda) = k\lambda^2$. Table 1 shows the correction magnitude.

Table 1: Jensen correction magnitude for selected parameter combinations.

λ	k	γ	Correction (%)
1.0	1.0	1.0	25.0
1.0	1.0	0.5	22.2
0.5	1.0	1.0	22.2
0.3	1.0	1.0	17.8
0.5	1.0	0.5	16.0
1.0	0.5	1.0	12.5
0.3	1.0	0.5	11.3
1.0	0.5	0.5	11.1

A simulation study with 10,000 replicates per scenario confirmed that the corrected formula reduces the effective rate estimation error from approximately 8% to 1% and produces slightly conservative power (**?@tbl-power-comparison**). During package development, this correction was needed for simulation-based power to agree with the analytical planning formula when event gaps were non-negligible.

Table 2: Power comparison: Jensen-corrected vs naive formula (10,000 replicates each). {#tbl-power-comparison} ### Broader parameter sweep

k	Gap (days)	n (corrected)	n (naive)	Power (corr.)	Power (naive)
0.5	20	436	422	0.9061	0.8992
1.0	20	712	684	0.9170	0.9033
0.5	30	454	436	0.9042	0.8934
1.0	30	740	700	0.9105	0.8988

The four scenarios in **?@tbl-power-comparison** fix $\lambda_1 = 0.4$ and vary only k and the gap duration. To characterize *when* the correction matters across a wider range of clinically relevant settings, we conducted a broader simulation sweep over three axes: the control event rate λ_1 , the dispersion k , and the event gap γ .

Parameter grid. The sweep covers $4 \times 4 \times 4 = 64$ scenarios defined by:

Table 3: Broad sweep parameter grid for Jensen correction impact study.

Parameter	Values	Clinical rationale
λ_1 (events/month)	0.3, 0.5, 1.0, 2.0	Heart failure hospitalizations (\$ 0.3) through COPD exacerbations (0.5) to MS relapses (\$1–2)
k (dispersion)	0.1, 0.3, 0.5, 1.0	Near-Poisson (0.1) to substantial patient heterogeneity (1.0); published NB trials typically report $k \in [0.3, 1.0]$
Gap (days)	10, 20, 30, 45	Short recovery (migraine) to extended gap (steroid taper in COPD exacerbations)

The remaining design parameters are held constant across all scenarios: $\lambda_2/\lambda_1 = 0.7$ (30% rate reduction), power = 90%, one-sided $\alpha = 0.025$, dropout rate = 0.1/12/month, maximum follow-up = 12 months, trial duration = 24 months, and piecewise accrual (rate R for 0–6 months, $2R$ for 6–12 months). These are fixed because they affect power symmetrically for both the corrected and naive designs, so they shift both power estimates together without meaningfully changing the *gap* between them. The three chosen axes (λ_1 , k , γ) correspond directly to the three terms in the correction factor:

$$1 - \frac{k\lambda_1\gamma}{(1 + \lambda_1\gamma)^2}$$

Rationale for disease-area calibration. The parameter ranges are informed by published recurrent event trials. Heart failure hospitalization trials such as PARADIGM-HF report control rates of approximately 0.3 events/month with dispersion near 0.5 (McMurray et al. 2014). COPD exacerbation trials (e.g., FLAME, IMPACT) report rates around 0.5–0.8/month with $k \in [0.5, 1.0]$ and commonly define a new exacerbation as requiring \$ 28 symptom – freedays [Wedzicha 2016, Lipson 2018]. Multiple sclerosis relapse trial observer rates of 1–2/year with moderate – to – high overdispersion [Hauser 2017, ocrelizumab]. At one extreme ($\lambda_1 = 0.3$, $k = 0.1$, gap = 10 days), the theoretical correction factor is below 0.1% and the naive formula suffices. At the other ($\lambda_1 = 2.0$, $k = 1.0$, gap = 45 days), it exceeds 10%, and the underpowering from the naive formula becomes practically meaningful.

Simulation protocol. For each scenario, we compute both the Jensen-corrected and naive sample sizes, simulate 3,500 trials at the corrected n , and extract results for the first n_{naive} subjects from the same simulated data. This paired design makes the power comparison very precise (SE ≈ 0.5 percentage points at 90% power). The aggregated results — scenario parameters, sample sizes, empirical power, paired power differences, and discordant pair counts — are saved for use in downstream analyses.

Results. Table 4 summarizes selected scenarios from the 64-scenario sweep, ordered by the theoretical correction factor. The corrected design achieves power within simulation error of the 90% target across all 64 scenarios (range: 0.896–0.924), while the naive design falls below 90% in the majority of scenarios where $k \geq 0.5$ and the gap exceeds 20 days.

Table 4: Selected scenarios from the broad sweep (3,500 replicates each). Rows are ordered from largest to smallest correction factor; the last two rows illustrate scenarios where the correction has no effect.

λ_1	k	Gap (days)	n (corr.)	n (naive)	Δn	Power (corr.)	Power (naive)	Δ Power (pp)
1.0	1.0	30	426	406	20	0.911	0.899	1.1
0.5	1.0	45	490	456	34	0.910	0.890	2.1
1.0	1.0	45	446	420	26	0.911	0.896	1.5
2.0	1.0	30	402	388	14	0.911	0.900	1.0
0.5	1.0	30	466	442	24	0.914	0.893	2.1
0.3	1.0	45	542	502	40	0.919	0.903	1.6
2.0	1.0	45	418	402	16	0.911	0.899	1.2
0.3	1.0	30	516	488	28	0.918	0.904	1.4
0.5	0.5	45	300	284	16	0.908	0.891	1.7
0.3	0.5	45	348	332	16	0.911	0.895	1.7
2.0	0.1	45	96	96	0	0.918	0.918	0.0
0.3	0.1	10	162	162	0	0.900	0.900	0.0

Three patterns emerge:

1. **The correction is driven by the $k \times \gamma$ interaction.** Dispersion $k \geq 0.5$ combined with gaps ≥ 30 days consistently produces 1–2 percentage points of underpowering from the naive formula, corresponding to 14–40 additional subjects (3–8% of total n).
2. **High event rates do not amplify the power gap as much as high dispersion.** At $k = 0.3$, even $\lambda_1 = 2.0$ with a 45-day gap shows only \$0.6 pp of underpowering. Conversely, at $k = 1.0$, $\lambda_1 = 0.5$ with a 30-day gap shows 2.1 pp — the largest gap in the sweep. This is because the correction factor $k\lambda\gamma/(1 + \lambda\gamma)^2$ saturates in λ but grows linearly in k .
3. **The correction is negligible when either k or γ is small.** All scenarios with $k = 0.1$ show $\Delta n \leq 4$ and $\Delta\text{Power} < 0.8$ pp. Similarly, all 10-day gap scenarios show $\Delta n \leq 10$ regardless of k .

These findings provide practical guidance: the Jensen correction is most valuable for trials in therapeutic areas with substantial patient heterogeneity ($k \geq 0.5$) and protocol-defined event gaps exceeding 20 days, such as COPD exacerbation studies with 28-day new-event windows or heart failure trials with extended hospitalization-free windows.

3 Group sequential design and sample size re-estimation

3.1 Integration with gsDesign

The `gsNBCalendar()` function creates a group sequential design by computing the statistical information at each planned analysis time, accounting for staggered enrollment and variable exposure. It returns an object that inherits from `gsDesign`, enabling use of spending functions, boundary summaries, and integer rounding from the `gsDesign` package.

3.2 Blinded and unblinded SSR

At interim analyses, nuisance parameters (k and λ_1) can be re-estimated to update the required sample size. Blinded SSR (Friede and Schmidli 2010; Schneider et al. 2013) fits a single NB model to pooled data and splits the estimated rate using the planned treatment effect ratio. Unblinded SSR uses treatment-specific estimates. In our simulations, blinded updates generally requested larger adapted sample sizes than unblinded updates, reflecting the conservatism of the pooled working model; when operationally acceptable, the unblinded route may therefore be preferable for sample efficiency. Both methods accept a `target_info` parameter from the group sequential design to account for inflation due to interim alpha spending.

The `calculate_blinded_info()` function includes a Poisson fallback threshold (matching `mutze_test()`) to guard against numerically unstable dispersion estimates that would produce extreme information values.

3.3 Spending function approach

For the simulation study, bounds are computed using spending functions with `usTime = lsTime = min(planned IF, actual IF)` at the interim. This ensures that neither efficacy nor futility spending exceeds the planned rate, even when more information than expected is available. Futility assessment is deferred until at least 30% of the planned information is observed.

4 Simulation study

4.1 Design

We evaluated SSR across 13 scenarios varying the control rate (λ_1), dispersion (k), accrual rate, and rate ratio ($RR = 0.7, 0.8, 0.9, 1.0$). The planned design targeted 90% power for $RR = 0.7$ with $k = 0.3$, $\lambda_1 = 0.5/\text{month}$, and accrual of 30/month over 12 months. Each non-null scenario used 3,600 simulations; the null scenario used 20,000.

Three strategies were compared: (1) no adaptation, (2) blinded SSR, and (3) unblinded SSR, all using a group sequential design with Hwang–Shih–DeCani spending functions.

4.2 Results

The key findings from the simulation study are:

- **Type I calibration:** Under the null ($RR = 1.0$, misspecified nuisance parameters), the main vignette configuration showed slight inflation at nominal one-sided 2.5%; a sensitivity rerun at one-sided 2.0% improved calibration.
- **Power recovery at $RR = 0.7$:** When the planned rate ratio was correct, both SSR methods recovered power from as low as 60% (no adaptation, worst case) to approximately 85–90%, by increasing the sample size up to $2 \times$ planned.
- **Diminishing returns at smaller effects:** For $RR = 0.8$ and $RR = 0.9$, the planned sample size (designed for $RR = 0.7$) was inherently insufficient. SSR compensated for nuisance parameter misspecification but could not overcome the smaller treatment effect.
- **Blinded vs unblinded:** Both methods recovered power, but blinded SSR generally requested larger adapted sample sizes than unblinded SSR because pooled nuisance updates were more

conservative. When operationally acceptable, unblinded SSR may therefore be preferable for efficiency.

5 Role of AI in package development

The **gsDesignNB** package was developed with substantial assistance from AI coding tools. The primary development environment was VS Code with GitHub Copilot and Cursor; models used over the course of development include Gemini 3.1, Claude Opus 4.6, and GPT-5.4x. The AI contributed to:

- **Code generation:** Implementation of simulation engines, data cutting functions, and the group sequential design interface.
- **Mathematical derivation:** The Jensen’s inequality correction was identified and derived through an interactive dialogue analyzing simulation discrepancies. The AI performed Monte Carlo verification and second-order Taylor expansion to pinpoint the bias source.
- **Documentation:** Vignettes, roxygen2 documentation, and notation consistency were developed collaboratively, with the AI drafting and the statistician reviewing for correctness.
- **Debugging:** Several subtle bugs (e.g., `toInteger()` overwriting enrollment with information-scaled values, positional argument mismatches in SSR calls, **gsDesign** collapsing futility bounds when `n.I` is provided directly) were identified through systematic diagnosis.
- **Simulation studies:** The multi-scenario SSR simulation and the 64-scenario Jensen correction sweep were designed, coded, and iteratively refined through conversation, with the statistician guiding the experimental design and the AI implementing it.

This workflow demonstrates that AI tools can accelerate statistical software development while requiring expert oversight for methodological correctness. The statistician’s role shifts toward design decisions, mathematical validation, and interpretation, while the AI handles implementation, debugging, and large-scale computation. AI was key to the productivity needed to complete this scope of research.

6 Discussion

The **gsDesignNB** package fills a gap in the R ecosystem for planning and adapting recurrent event trials with NB outcomes. While the **gscounts** package (Mütze et al. 2019) implements group sequential designs for NB data, **gsDesignNB** extends this with piecewise accrual, event gaps with Jensen’s correction, blinded and unblinded SSR, and comprehensive simulation utilities.

The Jensen’s inequality correction, while modest in many practical scenarios (1–8% effective rate reduction), becomes increasingly important as dispersion and event gap duration grow. The broader 64-scenario sweep (Table 4) confirms that the correction is driven primarily by the $k \times \gamma$ interaction: at $k \geq 0.5$ with gaps ≥ 30 days, the naive formula underpowers by 1–2 percentage points, whereas at $k = 0.1$ or gaps ≤ 10 days the correction adds negligibly to sample size. The extra subjects required by the correction range from 0 to 40 (0–8% of total), a modest cost for reliable power. The correction is principled (derived from the known Gamma frailty distribution) and conservative (slightly overpowering rather than underpowering).

The simulation study confirms that SSR is an effective tool for maintaining power when nuisance parameters are uncertain, provided the treatment effect assumption is correct. At the same time, the vignette configuration used here showed slight Type I inflation at nominal one-sided 2.5%,

with improved calibration after rebuilding at a smaller nominal α . This is best viewed as a design-specific operating-characteristic finding, consistent with the caution in Mütze et al. (2019), rather than as a universal correction rule.

For practice, the main operational message is that blinded SSR is conservative: it preserves masking but tends to request larger adapted sample sizes than unblinded updates. When charter and regulatory conditions allow unblinded nuisance updates, that route may be preferable for sample efficiency, provided the full operating characteristics are checked by simulation.

7 Software availability

The **gsDesignNB** package is available on GitHub at <https://github.com/keaven/gsDesignNB> with documentation at <https://keaven.github.io/gsDesignNB/>. The package requires $R \geq 4.1.0$ and depends on **gsDesign**, **data.table**, **simtrial**, and **MASS**.

References

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